

Rajalakshmi Engineering College

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Scan to verify results



2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Sesha is developing a weather monitoring system for a region with multiple weather stations. Each weather station collects temperature data hourly and stores it in a 2D array.

Write a program that can add the temperature data from two different weather stations to create a combined temperature record for the region.

Input Format

The first line of input consists of two space-separated integers N and M, representing the number of rows and columns of the matrices, respectively.

The next N lines consist of M space-separated integers, representing the values of the first matrix.

The following N lines consist of M space-separated integers, representing the values of the second matrix.

Output Format

The output prints the addition of the two matrices in N rows and M columns, representing the combined temperature record.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3 3

1 2 3

4 5 6

7 8 9

1 1 1

2 2 2

3 3 3

Output: 2 3 4

6 7 8

10 11 12

Answer

```
// You are using Java
import java.util.Scanner;
```

```
public class Main
{
```

```
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
```

```
        // Read matrix dimensions
```

```
        int N = scanner.nextInt();
```

```
        int M = scanner.nextInt();
```

```
        int[][] matrix1 = new int[N][M];
```

```
        int[][] matrix2 = new int[N][M];
```

```
        int[][] result = new int[N][M];
```

```
        // Read first matrix
```

```

    for (int i = 0; i < N; i++) {
        for (int j = 0; j < M; j++) {
            matrix1[i][j] = scanner.nextInt();
        }
    }

    // Read second matrix
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < M; j++) {
            matrix2[i][j] = scanner.nextInt();
        }
    }

    // Add both matrices
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < M; j++) {
            result[i][j] = matrix1[i][j] + matrix2[i][j];
        }
    }

    // Print the result matrix
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < M; j++) {
            System.out.print(result[i][j] + " ");
        }
        System.out.println(); // Move to next line after each row
    }

    scanner.close();
}

```

Status : Correct

Marks : 10/10